MyoDishAnalysis quick start: the watcher

Version 1.0.0-beta.5 · automatic analysis of all recordings of a folder · full description: README, *help MyoDishAnalysisWatch*, *mda-watch --help*

What it does

Every pass searches the raw folder and its subfolders for *.mdd* recordings and analyses those that are new or changed: every contraction of the whole recording, a summary per time bin (60 min) outside the stimulation protocols, and the protocols separately. Analysed recordings are skipped in the next passes unless the file changed. The raw folder is only read. MATLAB and Python write the same results and share the index.

Start

```
One pass:  MyoDishAnalysisWatch('/data/myodish/raw', '/data/myodish/results')
           mda-watch /data/myodish/raw /data/myodish/results                (Python)
Schedule:  MyoDishAnalysisWatch(RAW, RESULTS, 'schedule', 'daily 10:00')
           mda-watch RAW RESULTS --schedule "daily 10:00"     (Python, until Ctrl+C)
           'daily 10:00'  'weekly Sun 05:00'  'Mon,Thu 06:30'  'every 2 days 10:00'
           'every 120 min' (first pass now)  'every 6 h'
           MyoDishAnalysisWatch('status')   MyoDishAnalysisWatch('stop')
Options:   'rockerFilter', true   'compress', true   'fromDate', '2026-10-01'
           'dryRun', true (list only)   Python: --workers 6 (parallel)
```

With *'schedule'*, MATLAB checks for new recordings in the background at these times and stays free in between; the schedule ends with the MATLAB session. Without an open MATLAB: the scheduler of the operating system (cron, launchd, Windows Task Scheduler) with *mda-watch RAW RESULTS --quiet* or *matlab -batch "MyoDishAnalysisWatch('RAW', 'RESULTS', 'quiet', true)"*. Recordings still being written are skipped (changed in the last 10 min, or no *Recording stopped* in the log file yet).

Results folder

file	content
reports/mda_report_*.pdf	**start here**: analysis report of the pass (see below); mda_report_*.txt: short text version
<subfolder>/<name>_summary.csv	per channel and hour: contractions, capture, extra beats, amplitude, all parameters (mean, SD, n), comments of the log file, time without signal
<name>_contractions.csv(.gz), _pulses.csv	every contraction and every stimulus pulse
<name>_overview.csv	medians per minute: quick plots in Excel
<name>_protocols_*.csv	protocols (FFR, PRP, thresholds, refractory periods) and their results
<name>_channels.csv, _gaps.csv, _events.csv	status of each channel at the end (beating, not beating, removed, no slice), periods without signal, all log entries
mda_slices.csv	slice register: one row per slice from putting it in until it was taken out
mda_index.csv	one row per recording: status (ok, error, running, noLog), version, options, message

The analysis report

Page 1: all recordings of the pass × channels with capture (%) and flag codes. **Red** = check the analysis (it may not work, or technical faults put wrong values into the data): **I** inverted signal (sensor fault), **N** noise suspected (uncertain ≥ 25 % with extra + missed beats ≥ 15 %), **S** large extra beats (spikes), **D** delay pulse → onset > 150 ms, **F** pacing > 10 Hz (status channel misread). **Yellow** = slice quality (biology; you decide): **C** capture < 90 %, **X** extra beats > 10 %, **A** amplitude < 75 µN, **T** amplitude changes by > 30 % during the recording, **V** amplitude CV > 30 %. Grey: fewer than 50 contractions. Arrhythmically paced channels (detected, or option *arrhythmicChannels*): capture with ~, no C and V. Page 2: the list, with the channel label if given (e.g. *ch 7 (PMA 100 nM): noise index 54 (uncertain 98 %, extra 0 %, missed 55 %), amplitude change +660 % over 24 h*). Then traces of the red channels with the command that opens exactly this place in the GUI, with the settings of the watcher:

```
MyoDishAnalysisGUI('<results>/<name>_info.csv', 'channel', 3, 'window', [1200 1220])
mda-gui "<results>/<name>_info.csv" --channel 3 --window 1200 1220
```

MATLAB also prints these commands as links in the command window. Flags and thresholds: options *flagExtraBeats*, *flagUncertain*, *flagCapture*, … (Python: *flag_…*); *'report'*, *false* / *--no-report*: no PDF.

Good to know

- **After an update of MyoDishAnalysis** the earlier results are kept: only new and changed recordings are analysed (*'reanalyze'*, *'new'*, default). Because another version may give other values, a notice at the top of the report and in the command window says how many recordings were analysed with another version (or other options). To analyse them again with the current version: *'reanalyze'*, *'outdated'* (Python: *--reanalyze outdated*; can take hours). *'reanalyze'*, *'all'*: every recording again.
- Errors stop only the recording concerned (status *error* and message in the index); they are retried when the file or the version changes, or with *'retryErrors'*, *true*.
- One watcher per results folder (lock file *mda_watch.lock*). Labels per channel: *<name>_labels.csv* next to the recording (saved by the GUI). Same settings as in the GUI: *'settings'*, *file* (Advanced … → Save settings).